one-hour shot

- Software Engineering & Its Models
  - Characteristics of Software Engineering
  - Development Models
    - Build and fix Model
    - Waterfall Model
    - Iterative Enhancement Model
    - Prototyping Model
    - Spiral Model
    - RAD Model
  - Capability Maturity Model
    - Level 1 - Initial
    - Level 2 - Repeatable
    - Level 3 - Managed
    - Level 4 - Defined
    - Level 5 - Optmising

- Principles of Software Requirements Analysis
  - Functional Requirement
  - Non-Functional Requirement
  - Software Requirement Specification
    - Introduction

- Overall description
- Specific requirements
- Supporting information

▼ Requirement gathering Tools

- Record Review
- On-Site Observation
- Interview
- Questionaire

▼ Modeling System Architecture

▼ Elementary Modeling Technique

- Environment model
- Behavioral model

▼ Data Flow Diagram DFD

▼ Data Dictionary

▼ E-R Diagram

- Entity
- Relationship
- Cardinality & Optionality

▼ Software Prototyping and Specification

▼ Throw Away Prototype

▼ Evolutionary Prototype

▼ Software Metrics

▼ Size

▼ LOC

▼ Software Design

- ▼ Data Design
- ▼ Architectural Design
- ▼ Modular Design
- ▼ Cohesion
  - ▼ Concidential < Logical Procedural < Communication < Sequential < Functional
- ▼ Coupling
  - ▼ Content Coupling < Common Coupling < External coupling < Control Coupling < Stamp Coupling < Data Coupling
- ▼ Interface Design
  - ▼ Design of Human Computer Interface
- ▼ User Experience
- ▼ Design for Mobility

▼ Software Quality and Security

- ▼ Attributes of Software Quality
  - Auditability
  - Compatibility
  - Completeness
  - Consistency
  - Correctness
  - Feasibility
  - Modularity
  - Predictability
  - Robustness
  - Structuredness
  - Testability

- Traceability
- Understandability
- Verifiability

▼ Formal Technical Review

▼ Software Reliability

▼ Software reliability Models

▼ Software Quality Standards

▼ Capability Maturity Models

▼ Security Engineering

▼ Software Project Planning

▼ Types of Project metrics

▼ Product Metrics

▼ Process Metrics

▼ LOC

▼ Function Points

▼ External input | External Output | External user inquires | Internal logical files | External interface files

▼ Software Project Estimation

▼ Estimating Size

▼ Estimating Effort

▼ Estimating Schedule

▼ Estimating Cost

▼ Models for Estimation

▼ COCOMO Model

▼ Basic COCOMO

▼ Intermediate COCOMO

- Detailed COCOMO
- Classes of software projects
  - Organic
  - Semi-detached
  - Embedded
- Putnam's Model
- Statistical Model
- Automated Tools for Estimation

- Risk Management and Project Scheduling
  - Identification of Software Risks

    Basis for different types of Software Risk

    - Skills & Knowledge
    - Interface Modules
    - Poor Knowledge of tools
    - Programming Skills
    - Management Issue
  - Management of Risk
    - Risk Avoidance —> Risk Detection —> Risk Control —> Risk Recovery
  - Scheduling Methods
    - Work Breakdown Structure
    - Flow chart
      - Gantt Chart / Timeline Chart
      - PERT (Program Evaluation Review Technique)

- Software testing
  - Validation & Verification
  - Basic terms used in Testing

- Failure
- Fault
- Error
- Test cases

▼ Black Box Testing

▼ Boundary-value Analysis

▼ Equivalence Partitioning

▼ White Box testing

▼ Coverage Based testing

▼ Cyclomatic Complexity

▼ Mutation testing

▼ Levels of testing

▼ Unit testing

▼ Integration testing

▼ System Testing

▼ Debugging

▼ Life cycle of debugging

▼ Defect identification —> Defect Analysis —> Defect resolution

▼ Testing Tools

▼ Data Acquisition

▼ Static Measurement

▼ Dynamic Measurement

▼ Simulation

▼ Test Management

▼ Cross-functional tools

▼ Software Change Management

- Objectives of Software Change management process
  - Configuration identification
  - Configuration Control
  - Review
  - Status accounting
  - Auditing and reporting
- Version Control
- Change Control
- Auditing and Reporting

- Web Software Engineering
  - Characteristics of Web Application
  - Developement
  - Testing
  - Deployment
  - Issues of Management of Web based Projects
    - Review of Good Software Development Practices
      - Managing requirements
      - Managing Project
      - Configuration Management
      - Measurement
      - Risk management
    - Organization of Web Application teams
      - Webmaster
      - Application of Support Team
      - Content Development team
      - Web Publisher

- Development and Maintenance Issues
  - Configuration Management
  - Project Management
- Metrics
- Analysis
- Design and Construction
- Review and Testing

- Mobile Software Engineering
  - Transition from Design to Coding of Mobile Applications
    - Create a transition
    - Transition of Design to coding using wireframes
    - Prototyping mobile UI animations
    - Application Wire framing features
    - Animate layout changes using a transition
  - Elements of Mobile Application Development
    - Product Strategy
    - Comprehending what your application needs to deliver
    - User Interface design
    - Screen resolution
    - Graphics has to be appealing
    - User Experience Design
  - Mobile Device performance
    - Mobile application Size
    - Battery and Processor Usage
    - Social Sharing Option
    - Security

- Power Consumption
- Compatibility
- Simplicity
- Offline function
- Have a high performance

- Approach to development of mobile applications
  - Types of Mobile development Approaches
    - Native Application Development
    - Cross-Platform Application Development
    - Hybrid Application Development
    - Rapid Mobile Application Development
    - Progressive Web Application

- CASE Tools
  - Categories of CASE Tools
    - Upper CASE Tools
    - Lower CASE Tools
    - Integrated CASE Tools
  - Characteristics of CASE Tools
    - Standard methodology
    - Flexibility
    - String Integration
    - Support for reverse engineering
  - CASE Software Development Environment
  - CASE & Web engineering

- CASE Tools and requirement Engineering
  - Requirement elicitation
  - Software Analysis and Specification
- CASE Tools and Design & Implementation
- CASE Repository
  - Data
  - Process
  - Models
  - Rules/Constraints
- Software Testing
- Software Quality and CASE Tools
- Software Configuration Management
- Software Project Management and CASE Tools

- Advanced Topics in Software Engineering
  - Formal Methods
  - Use of Mathematics in Software Development
  - Limitation of Formal Specification using Formal Method
  - Cleanroom software Engineering
    - Increment Planning
    - Requirement gathering
    - Box Structure specification
    - Formal design
    - Correctness verification
    - Code generation, inspection, verification
    - Statistical test planning
    - Statistical use testing

- Conventional Software Engineering Models vs Cleanroom Software Engineering Model
- Software REUSE and its types
  - Application system reuse
  - Component Reuse
  - Function Reuse
- Component Based Software Engineering (CBSE) Process
  - Domain engineering
  - COmponent Based Development
    - Component qualification
    - Component Adaptation
      - White Box wrapping
      - Grey box wrapping
  - Reengineering
    - Existing —> System implementation —> System design —> System requirements —> Conceptual
- Software Process Improvement
  - Software Engineering Institute Capability Maturity Model Integrated
  - Software Process Optimization - First Time Right (FTR) Framework
  - Requirements related Optimization
    - Requirement traceability matrix
    - Metrics and SLA definition and sign-off.
    - NFR definition and sign-off
    - Business stakeholder engagement
    - Prototype demo
    - Design thinking Approach

- Architecture and Design Related Optimization
  - Patterns and best practices based design.
  - Tools, frameworks, package evaluation and fitment.
  - Standards and architecture principles definiton.
  - Optimal NFR and Integration design
  - Identification of automaton tools.
  - Feasibility validation PoCs
- Security Optimization
- Testing Optimization
  - Automation testing
  - Continous iterative testing
  - Testing metrics definition
  - Dashboard based quality monitoring
  - Sprint-wise early UAT.
- Development Optimization
  - Code checklist and guidance
  - Automated and Peer Review process
  - Code reusability
  - Performance driven development
  - Optimal code coverage
  - Quality gating criteria
    - developer-level code quality
    - automated local code quality analyzer
    - manual code review
      - integrated code review
    - Automated unit testability

- DevOps Optimization
  - Release management
  - Automated testing
  - Automated deployment
  - Continous build
  - Deployment pipeline
  - Source control manangement
  - Automated reporting and project health notification
- Infrastructure Optimization
  - Health Check monitoring setup
  - Proper capacity sizing
  - Automated alerts and notification
  - Monitoring dashboard
  - Availability reports
  - SLA based real time monitoring
  - Single Point of Failure avoidance
- Project Management Related Optimization
  - Appropriate skill training
  - Proper estimation and staffing
  - Reusable knowledge
  - Open communication channels
  - KT Planning for transition
  - Project metrics Dashboard
  - Sprint based Agile delivery
  - Continous Improvement model.
- Governance Optimization

- Code Governance
- Change request governance
- Defect fix governance
- Review governance
- Feature traceability governance
- Deployment governance
- Rollout governance.

- Emerging Trends in Software Engineering
  - Architecture brief details of emerging trends
    - Low Code and No Code Platform
    - Artificial Intelligence
      - Philosophy | Mathematics | Economics | Linguistic | Statistics | Neuroscience | Psychology | Computer Science| Control theory
    - BlockChain
    - Internet of Things
    - Augmented reality
    - Containerization
    - Continuous Delivery
    - Cloud Platform
    - Big Data
  - Other Emerging trends
    - Cross Platform development
    - Serverless Computing
    - Conversational Interface
- Introduction to UML
  - Classification of UML diagram

- Behavioral UML
  - Sequence Diagram
  - Activity Diagram
  - State Chart Diagram
- Structural UML
  - Class diagram
  - Package diagram
  - Component diagram
  - Deployment diagram
- Views Covered by UML diagram
  - Design view
  - Implementation view
  - Process view
  - Deployment view
- Structural UML Diagram

  Class | Interface | Object | Component | Use case
  - Class Diagram
    - Association
    - Dependency
    - Generalization
    - Realization
  - Object Diagram
  - Component diagram
  - Deployment diagram
  - Package diagram
- Behavioral UML diagram

- State Chart diagram
- Activity diagram
- Sequence diagram
- Use Case diagram
- Collaboration diagram

- Data Science for software engineers
  - Application of Data Science
    - Smart recommendations
    - Big Data Analysis
    - Web search
    - Data Analytics
    - Business Intelligence
    - Healthcare
    - Finance
    - Automobile
    - Supply Chain and Inventory Management
    - Telecom
  - Phases in Data Science Process
    - Requirements gathering and Data Discovery
    - Data Preparation Stage
    - Data exploration stage
    - Model development and prediction stage
    - Data visualization stage
  - Clustering Method
    - Collaborative filtering
  - Regression method

▼ Classification method